POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5033C | Semester V | Theory | 4 Credits

Electrical Engineering Materials

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5033C as Electrical Engineering Materials in Semester V for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kanpur’s Properties of Materials course and polytechnic electrical material curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Material testing results, engineering specifications, safety ratings and procurement plans must be based on approved laboratory data, certified material testing, current safety standards and competent professional review.

> **Verified source note:** Verified sources support conducting materials (free electron theory), insulating materials (dielectrics), magnetic materials, semiconductors and special purpose materials. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electrical Engineering Materials studies the properties and applications of substances used in the electrical industry. It develops the ability to analyze conducting, insulating, magnetic and semiconductor materials, evaluate their behavior under different electrical and thermal conditions, and understand the potential of special purpose materials (superconductors, nanomaterials) while respecting the technical and safety boundaries of material selection in engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic electrical engineering, engineering physics, chemistry and elementary mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the properties, classification and applications of conducting materials, including low and high resistivity alloys.
2. Explain the dielectric properties, classification and performance of insulating materials in solid, liquid and gaseous forms.
3. Explain the classification and properties of magnetic materials (soft and hard) and their applications in electrical machines.
4. Explain the fundamentals of semiconductor materials and the potential of special purpose materials like superconductors and nanomaterials.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the electrical and thermal properties of conducting materials and their industrial applications.	Understand
CO2	Explain the dielectric behavior, losses and breakdown mechanisms in insulating materials.	Understand
CO3	Explain the magnetic properties of materials and their use in transformers and rotating machines.	Understand
CO4	Explain the principles of semiconductors and the characteristics of special purpose engineering materials.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Conducting Materials	Free electron theory of metals; electrical and thermal conductivity; Wiedemann-Franz law; factors affecting resistivity; low resistivity materials (copper, aluminum); high resistivity materials (nichrome, constantan); superconductors.	Approx. 14 hours	CO1	Module 1
2	Insulating and Dielectric Materials	Dielectric constant; polarization mechanisms; dielectric loss and dissipation factor; dielectric strength and breakdown; solid insulators (mica, glass, ceramics); liquid insulators (transformer oil); gaseous insulators (air, SF_6).	Approx. 16 hours	CO2	Module 2
3	Magnetic Materials	Classification (Diamagnetism, Paramagnetism, Ferromagnetism); domain theory; Hysteresis loop; Eddy current loss; soft magnetic materials (CRGO steel, ferrites); hard magnetic materials (Alnico, rare-earth magnets); applications.	Approx. 16 hours	CO3	Module 3
4	Semiconductors and Special Materials	Intrinsic and extrinsic semiconductors; band theory basics; P-N junction; special purpose materials: superconductors (Meissner effect), nanomaterials (properties and potential), thermocouples, fuse materials, contact materials.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Conducting Materials

Free electron theory of metals; electrical and thermal conductivity; Wiedemann-Franz law; factors affecting resistivity; low resistivity materials (copper, aluminum); high resistivity materials (nichrome, constantan); superconductors.

CO1 · Approx. 14 hours

Insulating and Dielectric Materials

Dielectric constant; polarization mechanisms; dielectric loss and dissipation factor; dielectric strength and breakdown; solid insulators (mica, glass, ceramics); liquid insulators (transformer oil); gaseous insulators (air, SF_6).

CO2 · Approx. 16 hours

Magnetic Materials

Classification (Diamagnetism, Paramagnetism, Ferromagnetism); domain theory; Hysteresis loop; Eddy current loss; soft magnetic materials (CRGO steel, ferrites); hard magnetic materials (Alnico, rare-earth magnets); applications.

CO3 · Approx. 16 hours

Semiconductors and Special Materials

Intrinsic and extrinsic semiconductors; band theory basics; P-N junction; special purpose materials: superconductors (Meissner effect), nanomaterials (properties and potential), thermocouples, fuse materials, contact materials.

CO4 · Approx. 14 hours

MODULE 1

Conducting Materials

Free electron theory of metals; electrical and thermal conductivity; Wiedemann-Franz law; factors affecting resistivity; low resistivity materials (copper, aluminum); high resistivity materials (nichrome, constantan); superconductors.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

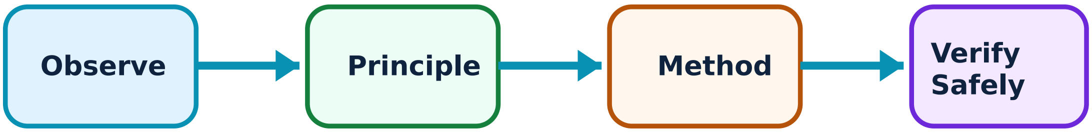
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Conducting Materials: identify the system, state its principle, follow the correct method and verify the result safely.

1. Conduction in metals−

Electrical conduction in metals is explained by the movement of free electrons. Conductivity depends on electron density and mobility. The Wiedemann-Franz law relates electrical and thermal conductivities. Resistivity increases with temperature due to increased electron-phonon scattering.

Study and application method

State the 'Wiedemann-Franz law'; list three factors that affect the resistivity of a conducting material.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State the 'Wiedemann-Franz law'; list three factors that affect the resistivity of a conducting material.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Resistivity is sensitive to impurities and mechanical stress. Do not use conducting materials for precision resistors without authorised calibration and temperature compensation.

2. Low and high resistivity materials−

Low resistivity materials like copper and aluminum are used for power transmission and windings. High resistivity materials like nichrome are used for heating elements, while constantan is used for precision resistors due to its low temperature coefficient.

Study and application method

Compare copper and aluminum as conductors for overhead transmission lines; list two applications of high-resistivity alloys.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare copper and aluminum as conductors for overhead transmission lines; list two applications of high-resistivity alloys.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Heating elements operate at high temperatures. Do not exceed the authorised current density to prevent premature oxidation or melting of the material.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Insulating and Dielectric Materials

Dielectric constant; polarization mechanisms; dielectric loss and dissipation factor; dielectric strength and breakdown; solid insulators (mica, glass, ceramics); liquid insulators (transformer oil); gaseous insulators (air, SF_6).

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

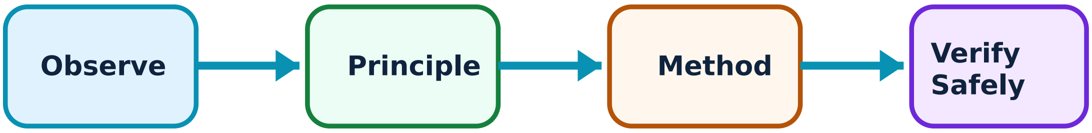
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Insulating and Dielectric Materials: identify the system, state its principle, follow the correct method and verify the result safely.

1. Dielectric properties and polarization−

Insulating materials (dielectrics) store electrical energy through polarization (electronic, ionic, dipolar). The dielectric constant measures this storage ability. Dielectric loss represents the energy dissipated as heat in an AC field, which must be minimized for efficient insulation.

Study and application method

Describe the four mechanisms of polarization in dielectrics; explain the significance of the 'dissipation factor' (tan delta).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the four mechanisms of polarization in dielectrics; explain the significance of the 'dissipation factor' (tan delta).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Dielectric loss increases with frequency and temperature. Do not use insulators in high-frequency applications without authorised loss-profile verification.

2. Breakdown and classification−

Dielectric strength is the maximum electric field a material can withstand before breakdown. Insulators are classified by their thermal stability (Class A to H). SF6 gas is widely used in high-voltage circuit breakers due to its excellent quenching properties.

Study and application method

List the factors affecting the dielectric strength of transformer oil; compare the properties of mica and ceramic insulators.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the factors affecting the dielectric strength of transformer oil; compare the properties of mica and ceramic insulators.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Dielectric breakdown can be catastrophic. Do not operate equipment above the authorised voltage ratings, and ensure regular testing of insulation resistance.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Magnetic Materials

Classification (Diamagnetism, Paramagnetism, Ferromagnetism); domain theory; Hysteresis loop; Eddy current loss; soft magnetic materials (CRGO steel, ferrites); hard magnetic materials (Alnico, rare-earth magnets); applications.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

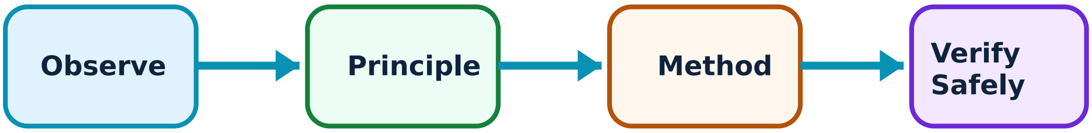
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Magnetic Materials: identify the system, state its principle, follow the correct method and verify the result safely.

1. Magnetism and losses−

Ferromagnetic materials exhibit strong magnetism due to domain alignment. Hysteresis and eddy current losses occur in AC applications. Silicon steel (CRGO) is used in transformer cores to minimize these losses and improve efficiency.

Study and application method

Sketch a typical Hysteresis loop and label the remanence and coercivity; explain how eddy current losses are reduced in transformer cores.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a typical Hysteresis loop and label the remanence and coercivity; explain how eddy current losses are reduced in transformer cores.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Magnetic materials can saturate, leading to non-linear behavior and heating. Do not exceed the authorised flux density in machine and transformer designs.

2. Soft and hard magnetic materials−

Soft magnetic materials (high permeability, low coercivity) are used for cores. Hard magnetic materials (high coercivity, high remanence) are used for permanent magnets. Ferrites are used in high-frequency applications due to their high resistivity.

Study and application method

Compare 'soft' and 'hard' magnetic materials in a conceptual table; list two applications of ferrite cores in electronics.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'soft' and 'hard' magnetic materials in a conceptual table; list two applications of ferrite cores in electronics.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Hard magnetic materials (like Neodymium) produce strong fields that can interfere with medical devices or electronic sensors. Follow authorised handling and storage protocols.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Semiconductors and Special Materials

Intrinsic and extrinsic semiconductors; band theory basics; P-N junction; special purpose materials: superconductors (Meissner effect), nanomaterials (properties and potential), thermocouples, fuse materials, contact materials.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

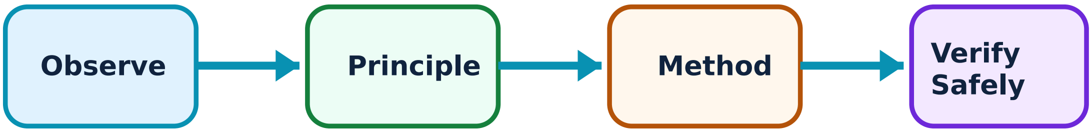
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Semiconductors and Special Materials: identify the system, state its principle, follow the correct method and verify the result safely.

1. Semiconductor fundamentals−

Semiconductors have conductivity between conductors and insulators. Band theory explains how doping (extrinsic) creates P-type or N-type materials. The P-N junction is the basic building block of diodes and transistors used in power electronics and control.

Study and application method

Explain the difference between 'intrinsic' and 'extrinsic' semiconductors using band diagrams; describe the formation of a 'depletion region' in a P-N junction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the difference between 'intrinsic' and 'extrinsic' semiconductors using band diagrams; describe the formation of a 'depletion region' in a P-N junction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Semiconductor properties are highly temperature-dependent. Do not operate devices without authorised thermal management (heat sinks/fans).

2. Special purpose materials−

Superconductors exhibit zero resistance and the Meissner effect (expulsion of magnetic fields) at low temperatures. Nanomaterials offer unique mechanical and electrical properties due to their high surface-to-volume ratio. Contact materials must resist arcing and oxidation.

Study and application method

Describe the 'Meissner effect' in superconductors; list two potential applications of nanomaterials in electrical engineering.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the 'Meissner effect' in superconductors; list two potential applications of nanomaterials in electrical engineering.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Superconductors require cryogenic cooling (e.g., liquid nitrogen/helium). Do not handle cryogenic fluids without authorised safety equipment and training.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Conducting Materials

Use the concept flow to revise observation, principle, method and verification.

Module 2: Insulating and Dielectric Materials

Use the concept flow to revise observation, principle, method and verification.

Module 3: Magnetic Materials

Use the concept flow to revise observation, principle, method and verification.

Module 4: Semiconductors and Special Materials

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare conductors, semiconductors and insulators in a conceptual table showing their band gap and conductivity.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the thermal classification chart for insulating materials (Class Y to C).

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the Hysteresis loops for a soft magnetic material and a hard magnetic material on the same axes.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the conducting and insulating materials used in a standard household electrical cable.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

List the properties required for an ideal fuse material and a high-voltage contact material.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Conducting Materials

Explain Conduction in metals and state one correct method, application or precaution.−

Electrical conduction in metals is explained by the movement of free electrons. Conductivity depends on electron density and mobility. The Wiedemann-Franz law relates electrical and thermal conductivities. Resistivity increases with temperature due to increased electron-phonon scattering.

Answer framework: State the 'Wiedemann-Franz law'; list three factors that affect the resistivity of a conducting material.

Important point: Resistivity is sensitive to impurities and mechanical stress. Do not use conducting materials for precision resistors without authorised calibration and temperature compensation.

Explain Low and high resistivity materials and state one correct method, application or precaution.−

Low resistivity materials like copper and aluminum are used for power transmission and windings. High resistivity materials like nichrome are used for heating elements, while constantan is used for precision resistors due to its low temperature coefficient.

Answer framework: Compare copper and aluminum as conductors for overhead transmission lines; list two applications of high-resistivity alloys.

Important point: Heating elements operate at high temperatures. Do not exceed the authorised current density to prevent premature oxidation or melting of the material.

Module 2 — Insulating and Dielectric Materials

Explain Dielectric properties and polarization and state one correct method, application or precaution.−

Insulating materials (dielectrics) store electrical energy through polarization (electronic, ionic, dipolar). The dielectric constant measures this storage ability. Dielectric loss represents the energy dissipated as heat in an AC field, which must be minimized for efficient insulation.

Answer framework: Describe the four mechanisms of polarization in dielectrics; explain the significance of the 'dissipation factor' (tan delta).

Important point: Dielectric loss increases with frequency and temperature. Do not use insulators in high-frequency applications without authorised loss-profile verification.

Explain Breakdown and classification and state one correct method, application or precaution.−

Dielectric strength is the maximum electric field a material can withstand before breakdown. Insulators are classified by their thermal stability (Class A to H). SF6 gas is widely used in high-voltage circuit breakers due to its excellent quenching properties.

Answer framework: List the factors affecting the dielectric strength of transformer oil; compare the properties of mica and ceramic insulators.

Important point: Dielectric breakdown can be catastrophic. Do not operate equipment above the authorised voltage ratings, and ensure regular testing of insulation resistance.

Module 3 — Magnetic Materials

Explain Magnetism and losses and state one correct method, application or precaution.–

Ferromagnetic materials exhibit strong magnetism due to domain alignment. Hysteresis and eddy current losses occur in AC applications. Silicon steel (CRGO) is used in transformer cores to minimize these losses and improve efficiency.

Answer framework: Sketch a typical Hysteresis loop and label the remanence and coercivity; explain how eddy current losses are reduced in transformer cores.

Important point: Magnetic materials can saturate, leading to non-linear behavior and heating. Do not exceed the authorised flux density in machine and transformer designs.

Explain Soft and hard magnetic materials and state one correct method, application or precaution.–

Soft magnetic materials (high permeability, low coercivity) are used for cores. Hard magnetic materials (high coercivity, high remanence) are used for permanent magnets. Ferrites are used in high-frequency applications due to their high resistivity.

Answer framework: Compare 'soft' and 'hard' magnetic materials in a conceptual table; list two applications of ferrite cores in electronics.

Important point: Hard magnetic materials (like Neodymium) produce strong fields that can interfere with medical devices or electronic sensors. Follow authorised handling and storage protocols.

Module 4 — Semiconductors and Special Materials

Explain Semiconductor fundamentals and state one correct method, application or precaution.–

Semiconductors have conductivity between conductors and insulators. Band theory explains how doping (extrinsic) creates P-type or N-type materials. The P-N junction is the basic building block of diodes and transistors used in power electronics and control.

Answer framework: Explain the difference between 'intrinsic' and 'extrinsic' semiconductors using band diagrams; describe the formation of a 'depletion region' in a P-N junction.

Important point: Semiconductor properties are highly temperature-dependent. Do not operate devices without authorised thermal management (heat sinks/fans).

Explain Special purpose materials and state one correct method, application or precaution.−

Superconductors exhibit zero resistance and the Meissner effect (expulsion of magnetic fields) at low temperatures. Nanomaterials offer unique mechanical and electrical properties due to their high surface-to-volume ratio. Contact materials must resist arcing and oxidation.

Answer framework: Describe the 'Meissner effect' in superconductors; list two potential applications of nanomaterials in electrical engineering.

Important point: Superconductors require cryogenic cooling (e.g., liquid nitrogen/helium). Do not handle cryogenic fluids without authorised safety equipment and training.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Conducting Materials.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Insulating and Dielectric Materials.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Magnetic Materials.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Semiconductors and Special Materials.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Free electron theory of metals; electrical and thermal conductivity; Wiedemann-Franz law; factors affecting resistivity; low resistivity

materials (copper, aluminum); high resistivity materials (nichrome, constantan); superconductors..

2. Develop a complete answer or practical plan for: Dielectric constant; polarization mechanisms; dielectric loss and dissipation factor; dielectric strength and breakdown; solid insulators (mica, glass, ceramics); liquid insulators (transformer oil); gaseous insulators (air, SF6)..
3. Develop a complete answer or practical plan for: Classification (Diamagnetism, Paramagnetism, Ferromagnetism); domain theory; Hysteresis loop; Eddy current loss; soft magnetic materials (CRGO steel, ferrites); hard magnetic materials (Alnico, rare-earth magnets); applications..
4. Develop a complete answer or practical plan for: Intrinsic and extrinsic semiconductors; band theory basics; P-N junction; special purpose materials: superconductors (Meissner effect), nanomaterials (properties and potential), thermocouples, fuse materials, contact materials..

LAST-MILE REVISION

Quick Revision

Module 1: Conducting Materials

Free electron theory of metals; electrical and thermal conductivity; Wiedemann-Franz law; factors affecting resistivity; low resistivity materials (copper, aluminum); high resistivity materials (nichrome, constantan); superconductors.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Insulating and Dielectric Materials

Dielectric constant; polarization mechanisms; dielectric loss and dissipation factor; dielectric strength and breakdown; solid insulators (mica, glass, ceramics); liquid insulators (transformer oil); gaseous insulators (air, SF6).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Magnetic Materials

Classification (Diamagnetism, Paramagnetism, Ferromagnetism); domain theory; Hysteresis loop; Eddy current loss; soft magnetic materials (CRGO steel, ferrites); hard magnetic materials (Alnico, rare-earth magnets); applications.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Semiconductors and Special Materials

Intrinsic and extrinsic semiconductors; band theory basics; P-N junction; special purpose materials: superconductors (Meissner effect), nanomaterials (properties and potential), thermocouples, fuse materials, contact materials.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Conduction in metals	Electrical conduction in metals is explained by the movement of free electrons.
Low and high resistivity materials	Low resistivity materials like copper and aluminum are used for power transmission and windings.
Dielectric properties and polarization	Insulating materials (dielectrics) store electrical energy through polarization (electronic, ionic, dipolar).
Breakdown and classification	Dielectric strength is the maximum electric field a material can withstand before breakdown.
Magnetism and losses	Ferromagnetic materials exhibit strong magnetism due to domain alignment.
Soft and hard magnetic materials	Soft magnetic materials (high permeability, low coercivity) are used for cores.
Semiconductor fundamentals	Semiconductors have conductivity between conductors and insulators.
Special purpose materials	Superconductors exhibit zero resistance and the Meissner effect (expulsion of magnetic fields) at low temperatures.

LEARNING RESOURCES

References

Recommended electrical-materials references

1. A. J. Dekker, Electrical Engineering Materials.
2. S. P. Seth, A Course in Electrical Engineering Materials.
3. C. S. Indulkar and S. Thiruvengadam, An Introduction to Electrical Engineering Materials.
4. R. K. Rajput, A Course in Electrical Engineering Materials.

Approved public electrical-materials sources

- https://nptel.ac.in/courses/113104090
- https://nptel.ac.in/courses/113104096

These sources provide the verified study basis while official Course 5033C detail is unavailable; they do not replace official material testing, authorised engineering specifications, certified safety standards or authorised industrial plans.